



哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

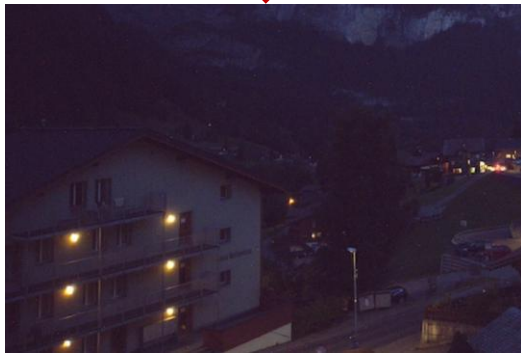
直面复原的数据之殇： 自监督图像与视频复原技术探索

汇报人：张志路

哈尔滨工业大学



超分辨率



去噪



去模糊



动态范围增强

低质量图像/视频

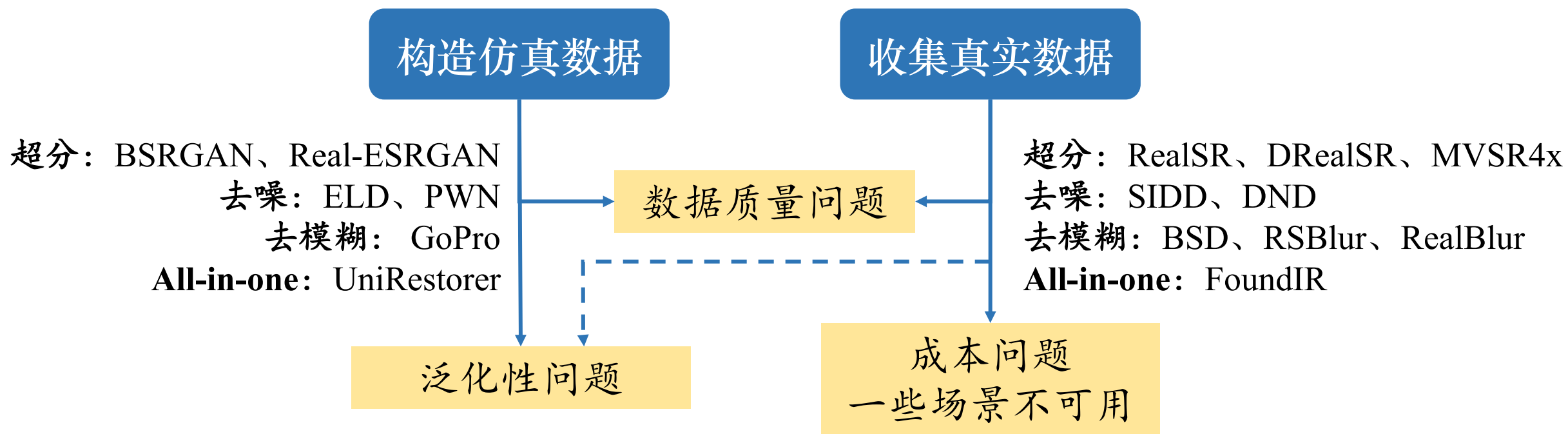


复原与增强模型



高质量图像/视频

■ 有监督复原模型的数据之殇

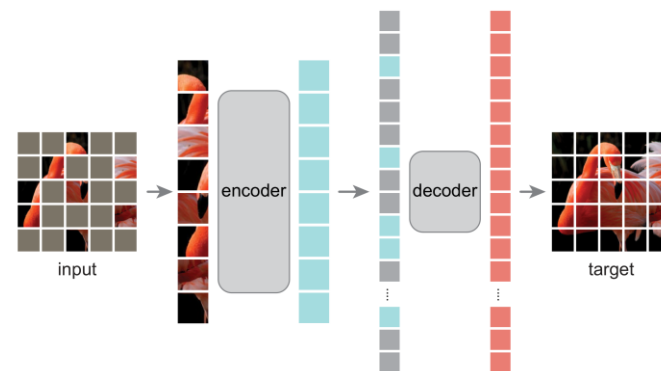
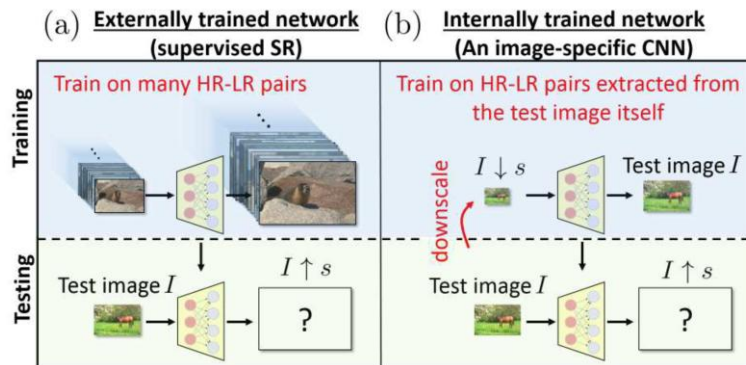
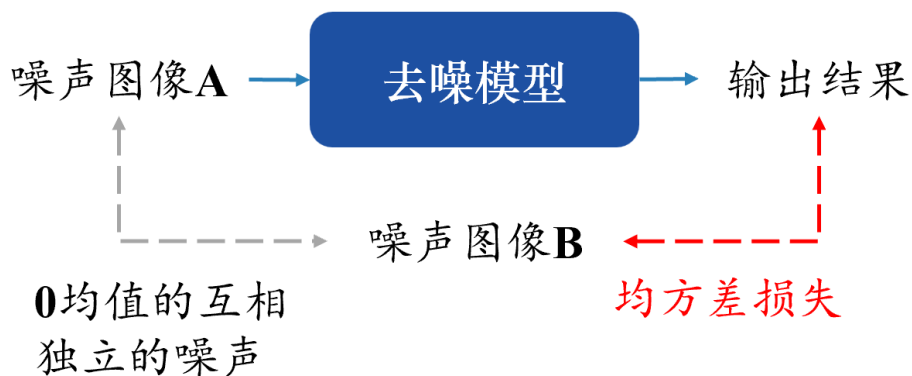
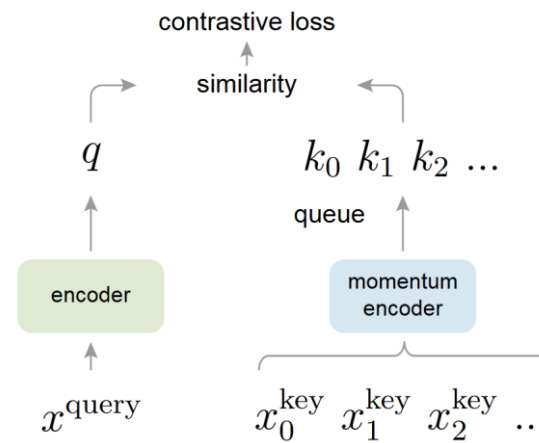
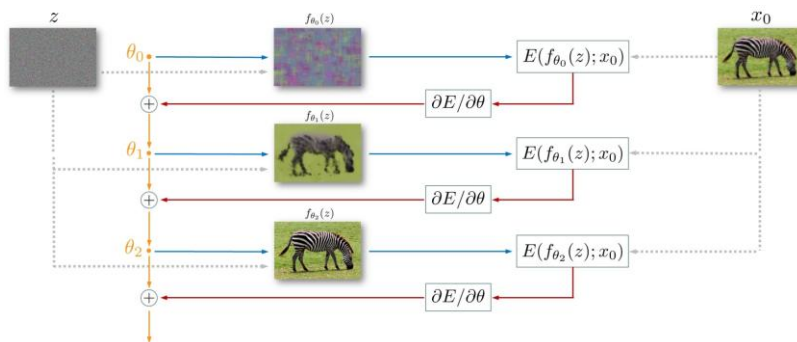


背景

自监督学习

■ 无需配对数据，利用数据内部信息或网络先验构造训练对

- 数据成本低
- 可直接面向真实场景
- 可测试时自适应
- Scaling Up空间大



- **可用：能在真实数据上训练**
- **效果：有助于提升现有复原模型效果**
- **泛化：能在一定规模数据集上训练，而非只能单场景优化**

有应用价值，可落地

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多曝光图像复原

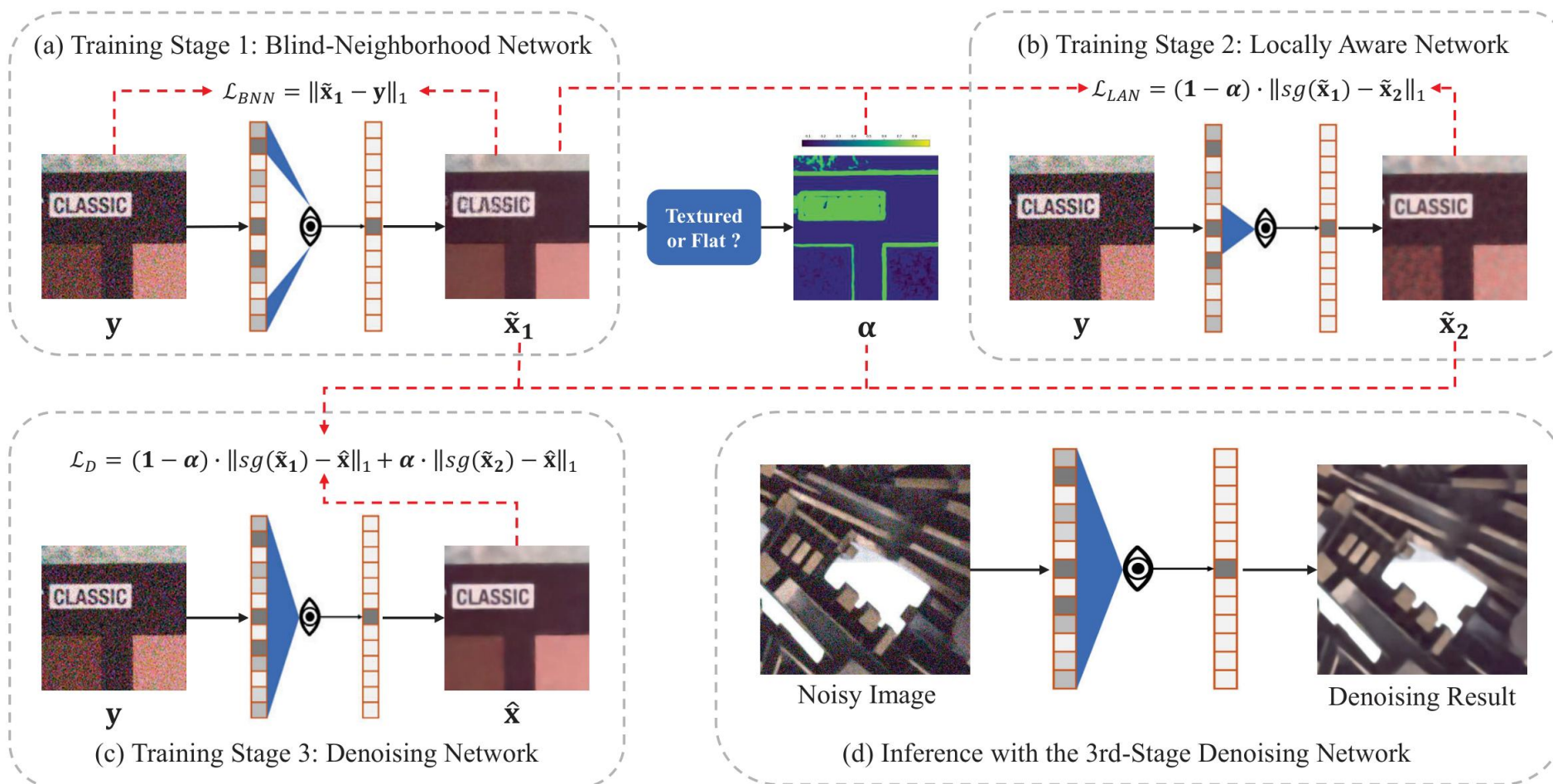
4

端侧落地情况



一、单图复原

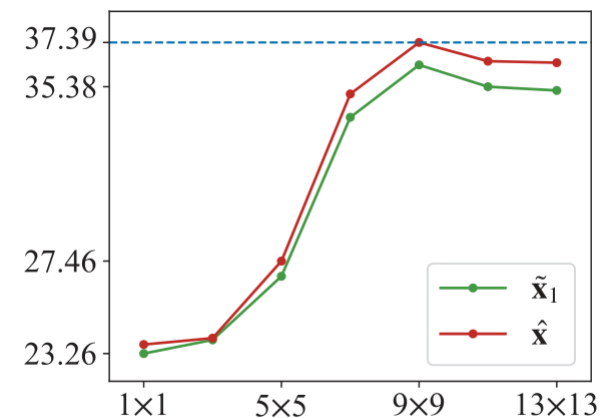
■ 真实RGB图像自监督去噪方法SASL



■ 真实RGB图像自监督去噪方法SASL

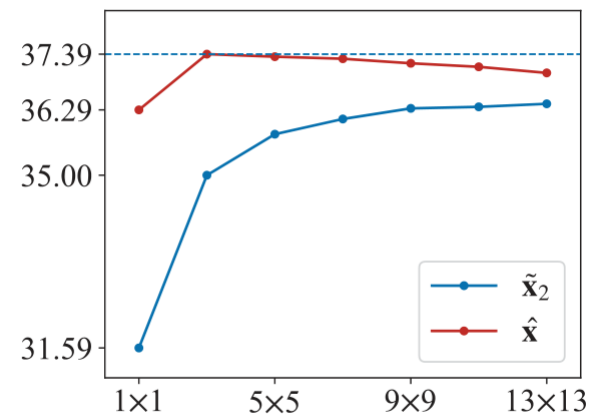
	Method	SIDD Validation PSNR↑ / SSIM↑ / LPIPS↓	SIDD Benchmark PSNR↑ / SSIM↑	DND Benchmark PSNR↑ / SSIM↑
Non-learning based	BM3D [11]	25.71 / 0.576 / 0.657	25.65 / 0.685	34.51 / 0.851
	WNNM [14]	26.05 / 0.592 / 0.635	25.78 / 0.809	34.67 / 0.865
Supervised (Synthetic pairs)	DnCNN [50]	26.21 / 0.604 / 0.712	26.25 / 0.599	32.43 / 0.790
	CBDNet [15]	33.07 / 0.863 / 0.288	33.28 / 0.868	38.05 / 0.942
	Zhou <i>et al.</i> [57]	33.96 / 0.899 / 0.258	34.00 / 0.898	38.40 / 0.945
Supervised (Real pairs)	DnCNN [50]	37.73 / 0.943 / 0.245	37.61 / 0.941	38.73 / 0.945
	Baseline, N2C [39]	38.98 / 0.954 / 0.201	38.92 / 0.953	39.37 / 0.954
	VDN [46]	39.29 / 0.956 / 0.208	39.26 / 0.955	39.38 / 0.952
	Restormer [48]	39.93 / 0.960 / 0.198	40.02 / 0.960	40.03 / 0.956
Unpaired	G CBD [8]	-	-	35.58 / 0.922
	UIDNet [16]	-	32.48 / 0.897	-
	C2N [18]	35.36 / <u>0.932</u> / <u>0.192</u>	35.35 / 0.937	37.28 / 0.924
	Wu <i>et al.</i> [44]	-	-	37.93 / <u>0.937</u>
Self-Supervised	Noise2Void [24]	27.48 / 0.664 / 0.592	27.68 / 0.668	-
	Noise2Self [3]	29.94 / 0.782 / 0.556	29.56 / 0.808	-
	NAC [45]	-	-	36.20 / 0.925
	R2R [36]	-	34.78 / 0.898	-
	CVF-SID [35]	34.15 / 0.911 / 0.423	34.71 / 0.917	36.50 / 0.924
	AP-BSN+R ³ [27]	<u>36.74</u> / 0.934 / 0.281	<u>36.91</u> / 0.931	<u>38.09</u> / <u>0.937</u>
	Ours	37.39 / 0.934 / 0.176	37.41 / <u>0.934</u>	38.18 / 0.938

PSNR(dB)



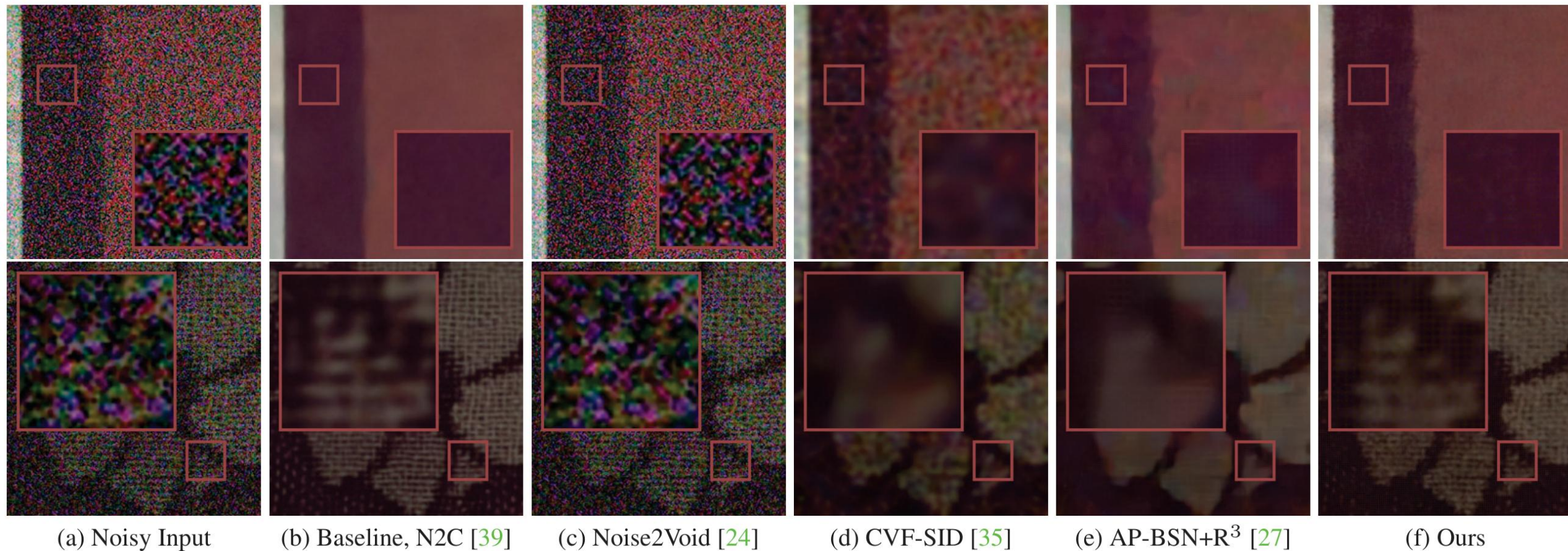
(a) Blind-neighborhood size.

PSNR(dB)

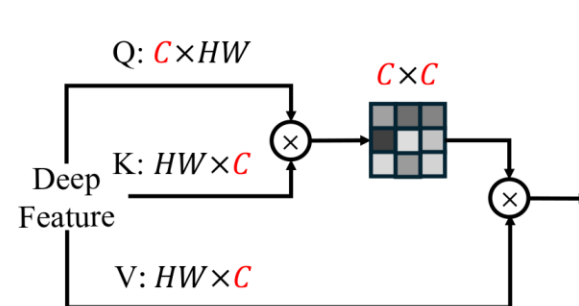
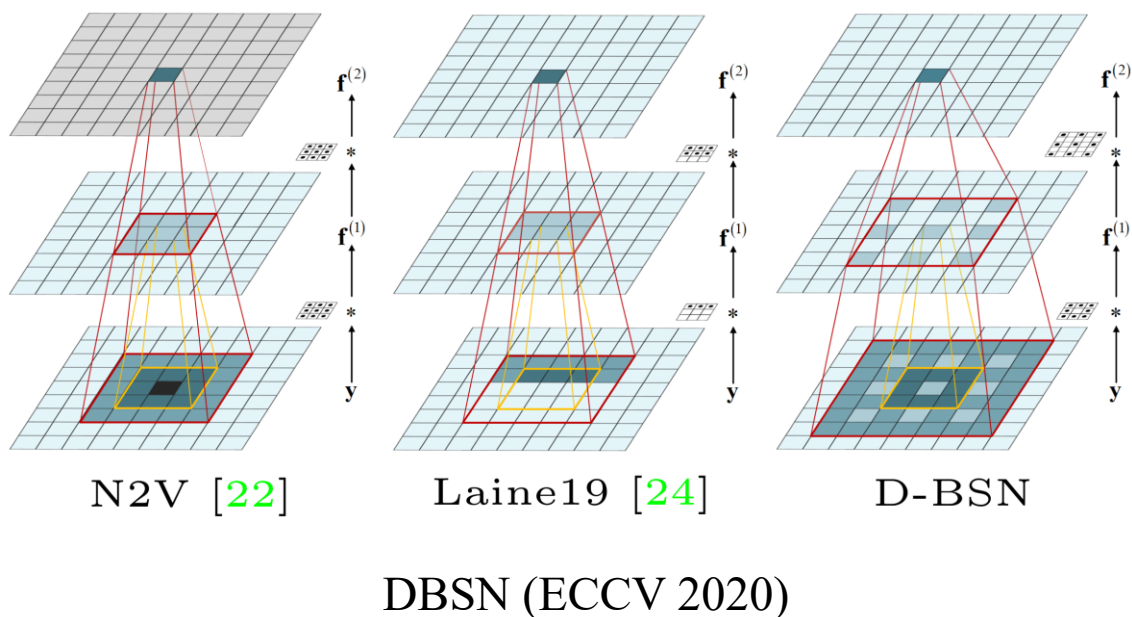


(b) Local receptive size.

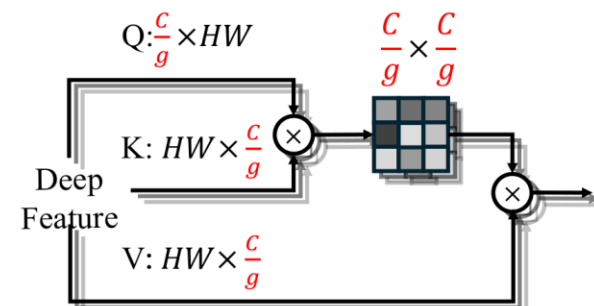
■ 真实RGB图像自监督去噪方法SASL



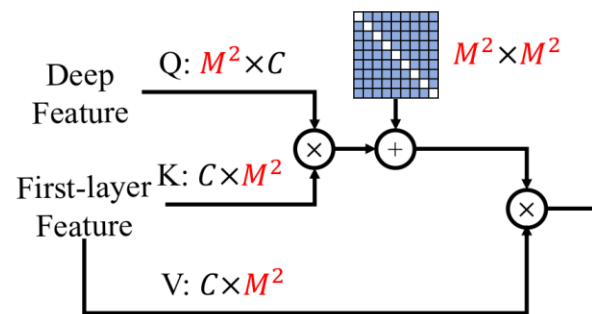
■ 基于Transformer的盲点网络TBSN



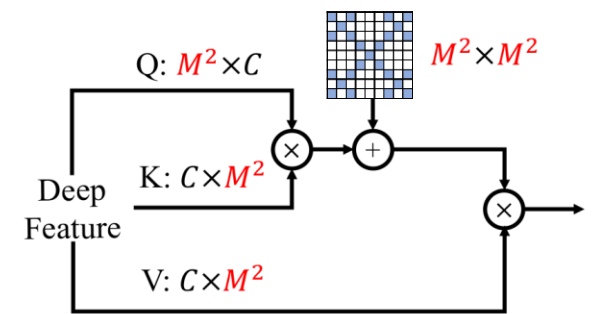
(a) Channel attention in LGBP



(b) G-CSA in our TBSN



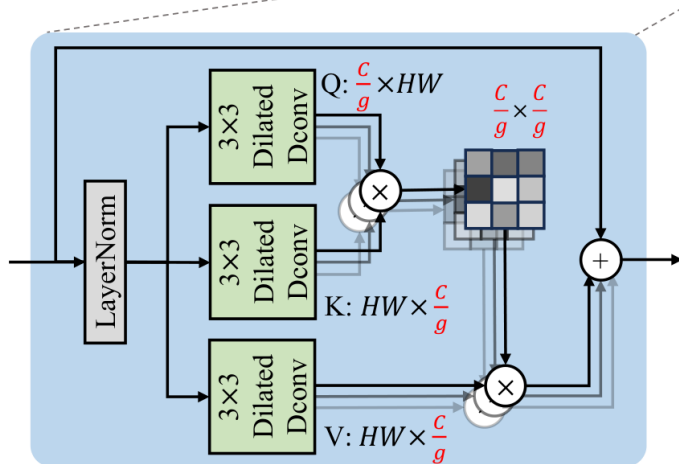
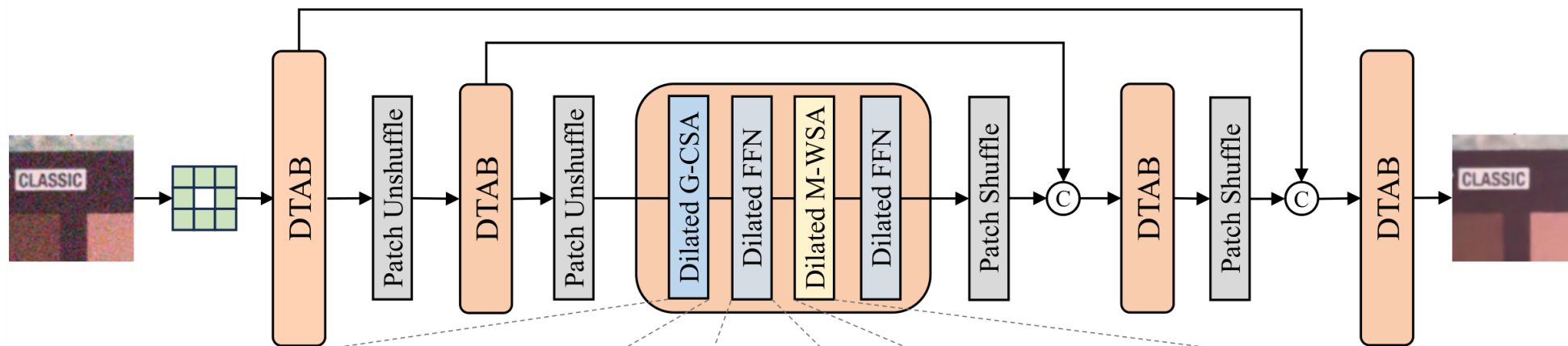
(c) Window attention in SwinIA



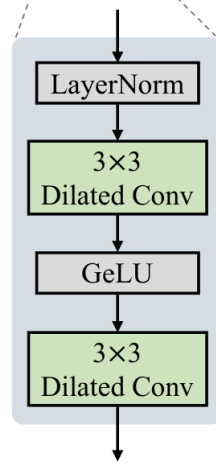
(d) M-WSA in our TBSN

■ 基于Transformer的盲点网络TBSN

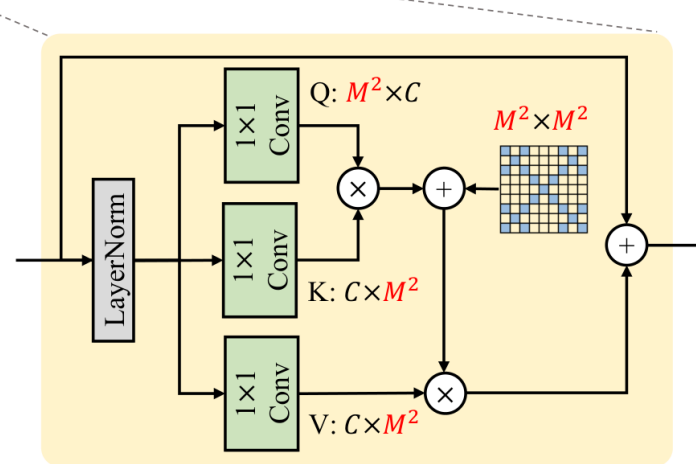
(a) Our Transformer-Based Blind-Spot Network (TBSN)



(b) Dilated G-CSA in DTAB

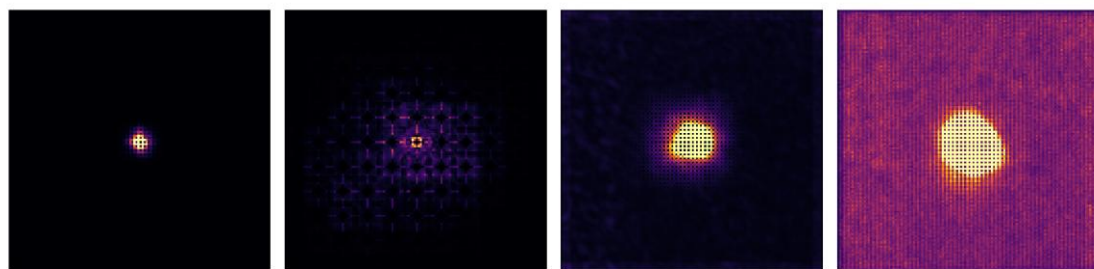


(c) Dilated FFN



(d) Dilated M-WSA in DTAB

■ 基于Transformer的盲点网络TBSN



AP-BSN

LG-BPN

PUCA

TBSN (Ours)

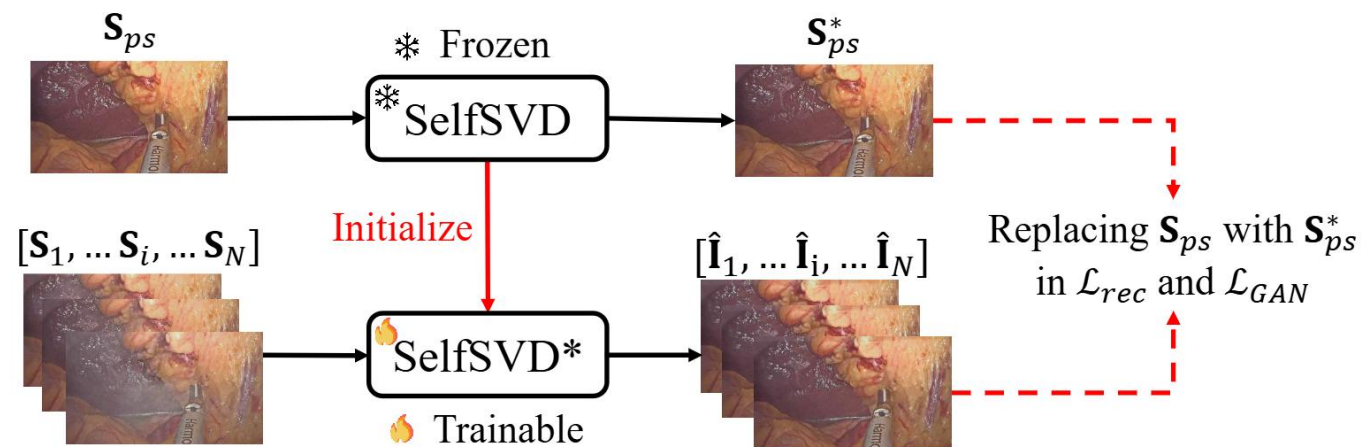
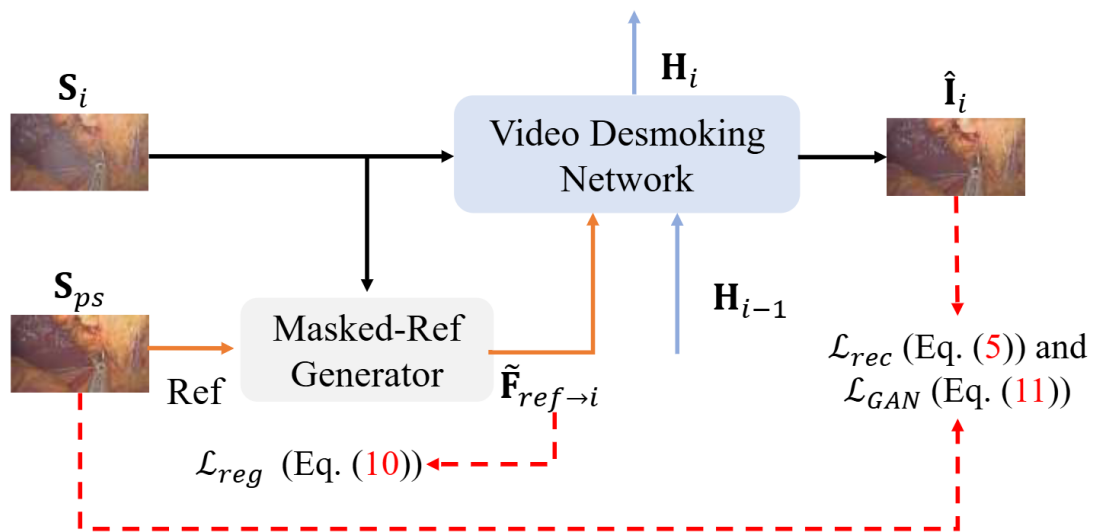
Method	SASL	LG-BPN	PUCA	TBSN	TBSN2UNet
PSNR (dB)	37.41	37.28	37.54	37.78	37.79
#Param (M)	1.08	6.61	12.78	12.97	1.08
#FLOPs (G)	35.0	6699.6	2644.2	5463.9	35.0
Time (ms)	4.8	5229.4	444.7	1032.1	4.8

	Method	SIDD Bench PSNR / SSIM	DND Bench PSNR / SSIM
Non-learning Based	BM3D	25.65 / 0.685	34.51 / 0.851
	WNNM	25.78 / 0.809	34.67 / 0.865
Supervised (Synthetic pairs)	DnCNN	26.25 / 0.599	32.43 / 0.790
	CBDNet	33.28 / 0.868	38.05 / 0.942
	Zhou <i>et al.</i>	34.00 / 0.898	38.40 / 0.945
Supervised (Real pairs)	DnCNN	37.61 / 0.941	38.73 / 0.945
	VDN	39.26 / 0.955	39.38 / 0.952
	Restormer	40.02 / 0.960	40.03 / 0.956
	NAFNet	40.30 / 0.961	- / -
Unpaired	GCBD	-	35.58 / 0.922
	UIDNet	32.48 / 0.897	-
	C2N	35.35 / 0.937	37.28 / 0.924
	DBSN	-	37.93 / 0.937
Self-Supervised	Noise2Void	27.68 / 0.668	-
	Noise2Self	29.56 / 0.808	-
	NAC	-	36.20 / 0.925
	R2R	34.78 / 0.898	-
	CVF-SID	34.71 / 0.917	36.50 / 0.924
	AP-BSN	36.91 / 0.931	38.09 / 0.937
	SASL	37.41 / 0.934	38.18 / 0.938
	LG-BPN	37.28 / 0.936	38.43 / 0.942
	PUCA	37.54 / 0.936	38.83 / 0.942
	AT-BSN	37.78 / 0.944	38.68 / 0.942
Self-Supervised (Ours)	TBSN	<u>37.78</u> / 0.940	39.08 / 0.945
	TBSN2UNet	37.79 / <u>0.940</u>	<u>39.01</u> / 0.945



二、视频复原

■ 腹腔镜视频自监督去烟雾方法SelfSVD

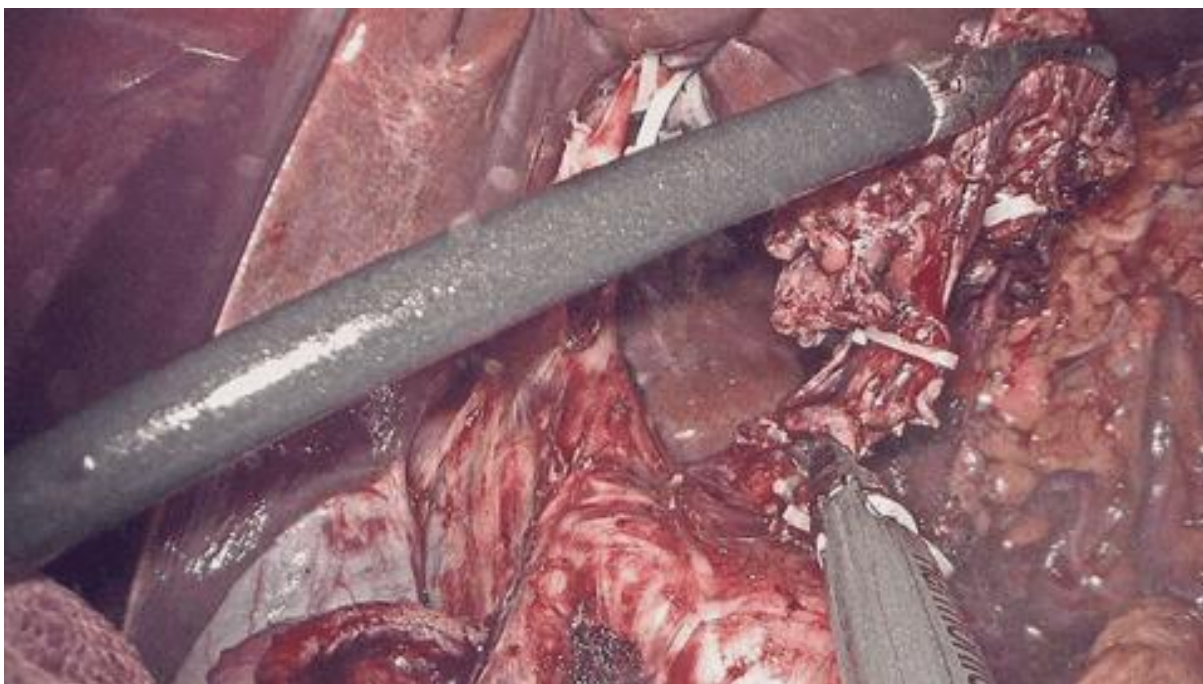


Self-Supervised Video Desmoking for Laparoscopic Surgery. ECCV 2024 Oral.

■ 腹腔镜视频自监督去烟雾方法SelfSVD

	Methods	PSNR↑	SSIM↑	FADE↓	NIQE↓	PI↓	#FLOPs (G)	#Time (s)	#Params (M)
Unsupervised Image Processing	PSD [9]	14.69	0.3722	0.1715	5.44	4.37	905	0.31	31.11
	DCP [21]	17.77	0.5582	0.3217	5.53	4.25	-	0.08	-
Unpaired Image Processing	DCP-Pixel2Pixel [50]	19.65	0.5079	0.5509	6.30	4.50	195	0.02	54.40
	DistentGAN [52]	21.51	0.6037	0.6135	4.95	3.60	1572	0.22	11.38
	Desmoke-LAP [43]	22.52	0.6170	0.5386	4.75	3.93	1571	0.15	11.38
	RefineNet [70]	22.87	0.6177	0.5749	6.93	4.13	1780	0.34	0.85
Self-Supervised Image Processing	UHD [63]	20.93	0.5792	1.3573	6.26	6.18	123	0.27	34.55
	MSDesmoking [59]	21.53	0.5997	1.1290	5.20	5.19	605	0.04	8.80
	Wang <i>et al.</i> [60]	22.65	0.6078	1.1202	7.63	7.11	6584	28.20	3.19
	MSBDN [11]	22.69	0.6093	0.8995	6.07	5.40	779	0.29	31.35
	AODNet [29]	23.03	0.6116	1.0417	5.96	4.79	4	0.08	0.002
	DADNet [20]	23.06	0.6143	0.4948	5.29	4.08	198	0.03	0.85
	Dehamer [19]	23.13	0.6184	1.1508	6.93	6.36	1564	0.41	132.45
Self-Supervised Video Processing	BasicVSR [6]	23.00	0.6168	0.5609	5.60	4.13	831	0.13	6.29
	BasicVSR++ [7]	23.35	0.6196	0.5665	5.50	3.90	1197	0.19	9.76
	MAPNet [64]	23.28	0.6152	0.9632	5.53	5.43	261	0.20	28.75
	(Ours) SelfSVD-S	23.84	0.6183	0.4787	5.04	3.94	169	0.03	1.92
	(Ours) SelfSVD*-S	24.00	0.6209	0.4404	4.87	3.92	169	0.03	1.92
	(Ours) SelfSVD	24.23	0.6216	0.4626	4.85	3.87	996	0.18	15.58
	(Ours) SelfSVD*	24.58	0.6279	0.4193	4.72	3.86	996	0.18	15.58

■ 腹腔镜视频自监督去烟雾方法SelfSVD

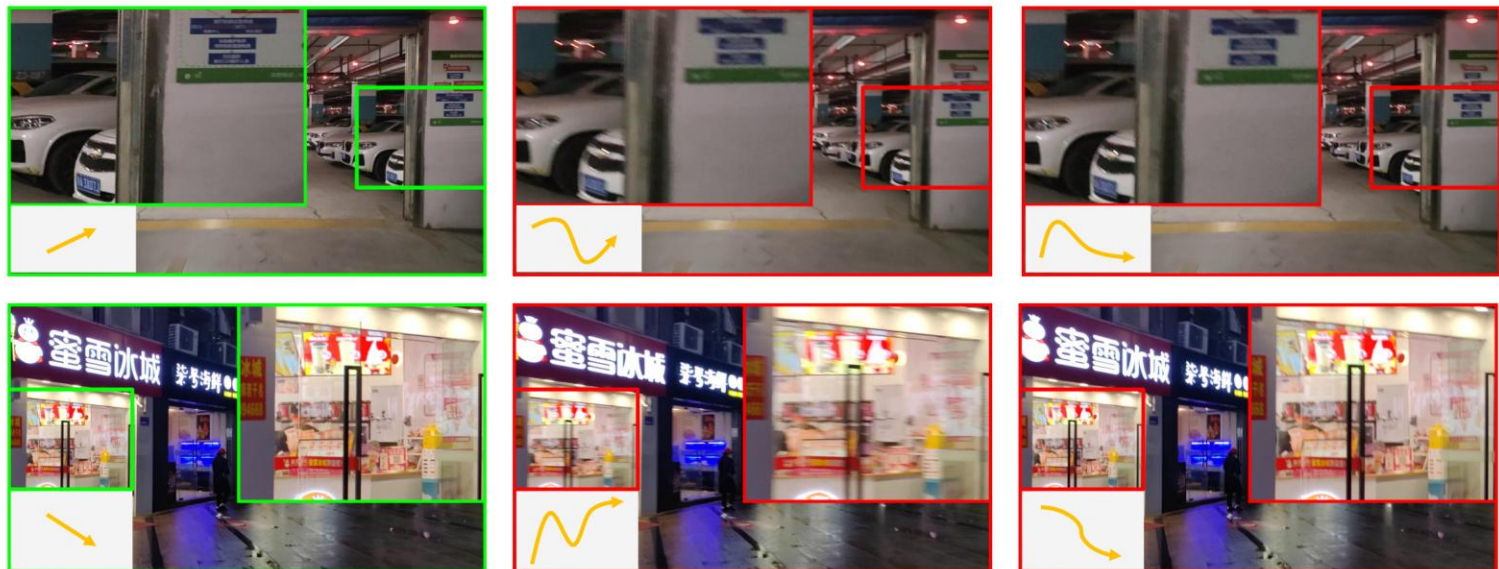


输入视频

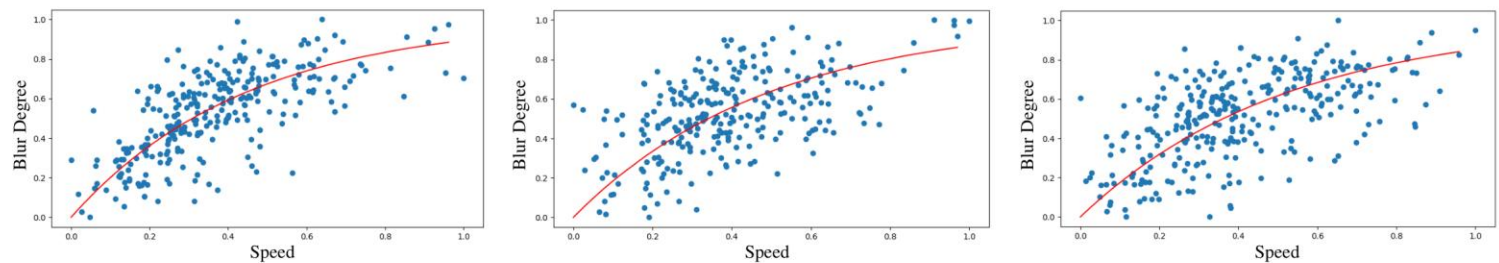


输出视频

■ 手机拍摄视频自监督去模糊方法SelfHVD



(a) Effect of smartphone motion trajectory on image blur



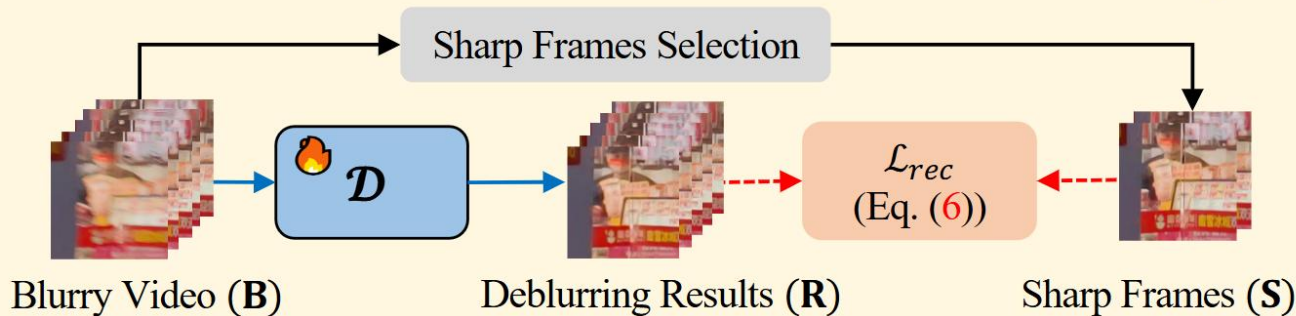
(b) Effect of smartphone motion speed on image blur

Brand	Model	Release Year	OIS	Sharp Frame Ratio
Huawei	Mate 50	2022	✓	33.33%
	Mate 60	2023	✓	32.50%
	Mate 70	2024	✓	38.85%
Xiaomi	Xiaomi 13	2022	✓	34.34%
	Xiaomi 14	2023	✓	31.91%
	Xiaomi 15	2024	✓	33.68%
Apple	iPhone 14	2022	✓	30.00%
	iPhone 15	2023	✓	32.55%
	iPhone 16	2024	✓	33.11%

■ 手机拍摄视频自监督去模糊方法SelfHVD

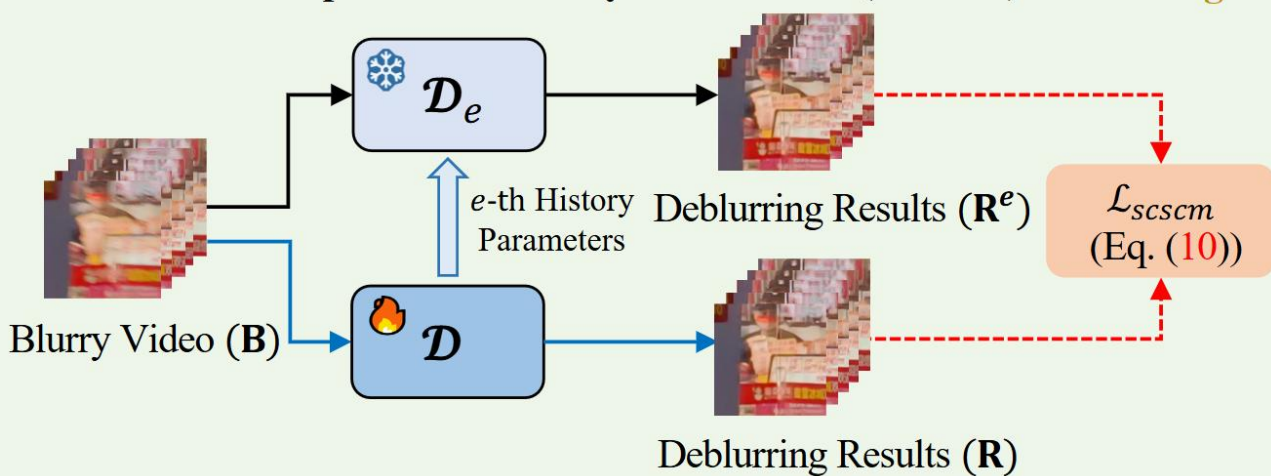
Selecting Sharp Frames as Supervision

Stage 1



Self-Constrained Spatial Consistency Maintenance (SCSCM)

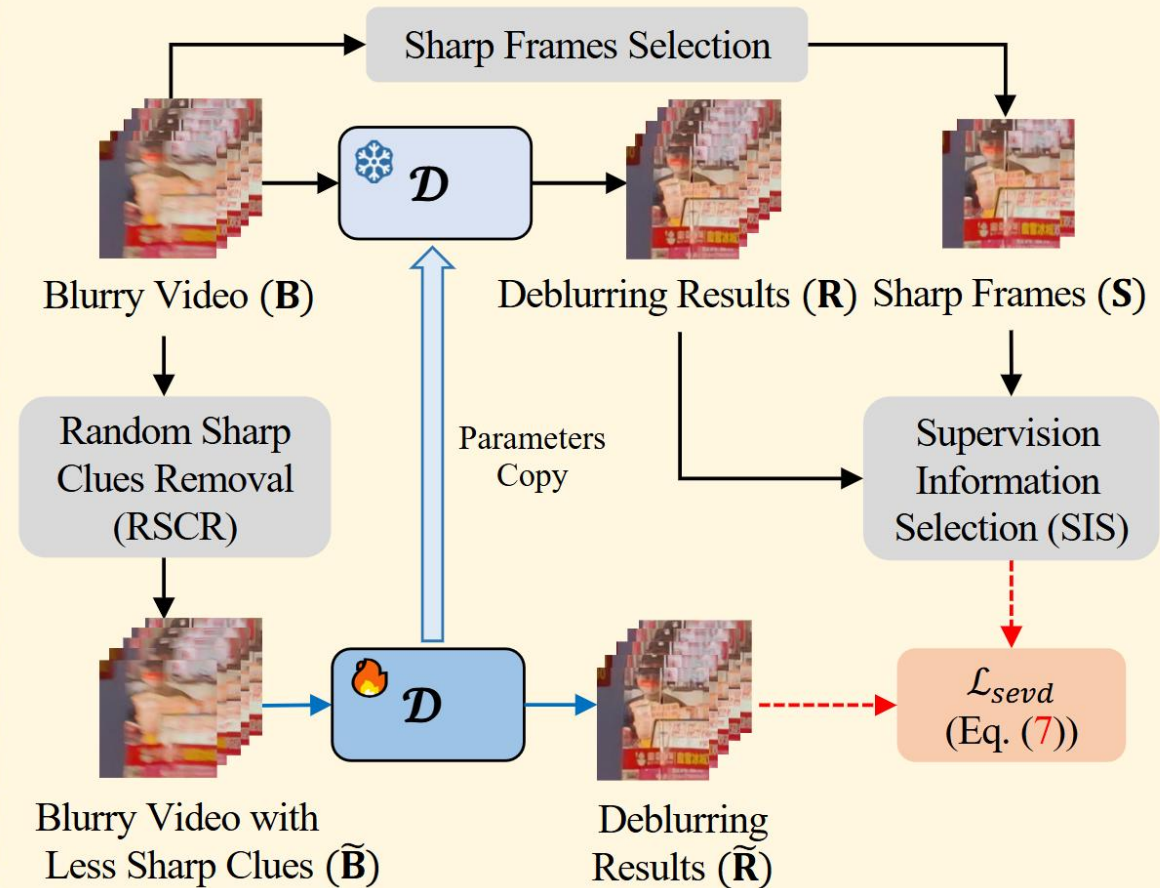
Stage 2



Spatial Consistency Maintenance

Self-Enhanced Video Deblurring (SEVD)

Stage 2



Video Restoration

■ 手机拍摄视频自监督去模糊方法SelfHVD

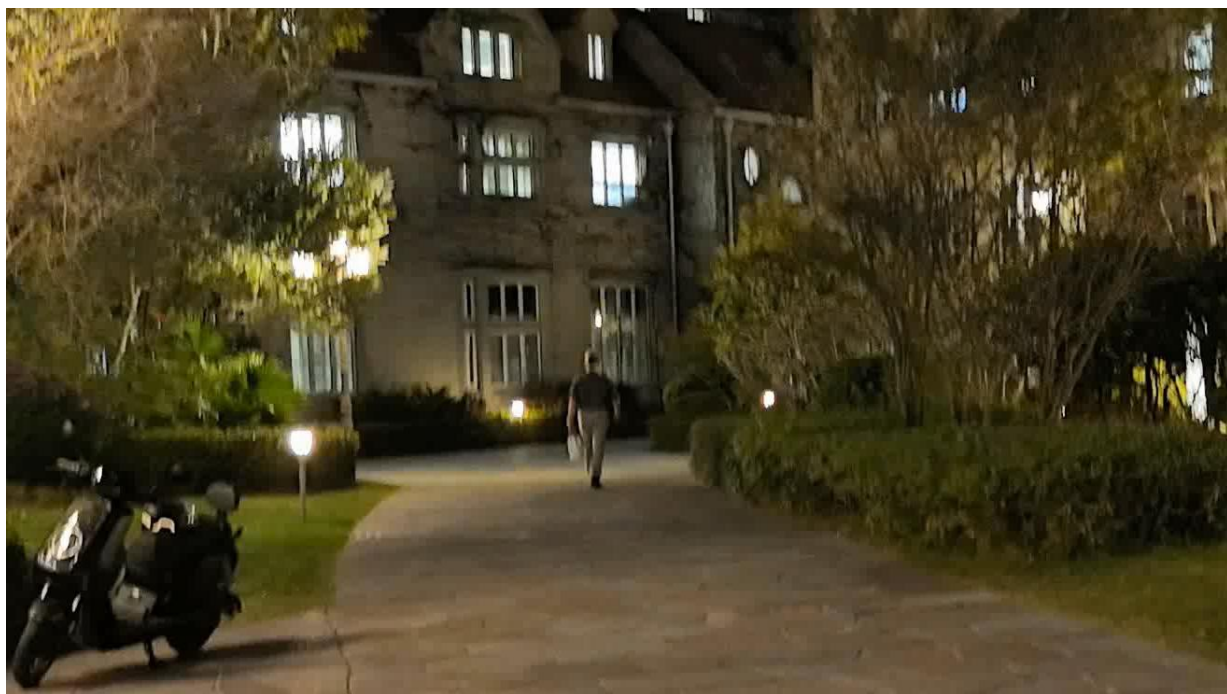
Table 1. Quantitative comparison on the synthetic GoProShake and real-world HVD datasets. ‘*Network*’ in ‘SelfHVD_{*Network*}’ denotes the deblurring network we use, where ESTRNN [64] is also adopted by Ren *et al.* [55] and DaDeblur [12].

Methods		GoProShake PSNR↑ / SSIM↑	HVD-Huawei MUSIQ↑ / MANIQA↑	HVD-Xiaomi MUSIQ↑ / MANIQA↑	HVD-iPhone MUSIQ↑ / MANIQA↑
Fully-Supervised	IFIRNN [34]	34.66 / 0.9448	24.1043 / 0.1916	29.7281 / 0.2212	22.3710 / 0.2535
	ESTRNN [64]	34.19 / 0.9369	24.0383 / 0.1917	29.4117 / 0.2193	21.6487 / 0.2506
	RVRT [23]	37.02 / 0.9473	24.9269 / 0.1923	30.2976 / 0.2215	22.4535 / 0.2546
	BasicVSR++ [3]	37.99 / 0.9683	25.2499 / 0.2006	30.0775 / 0.2235	22.6709 / 0.2564
Self-Supervised	Ren <i>et al.</i> [55]	25.05 / 0.7428	22.5433 / 0.1771	22.7757 / 0.2193	20.2299 / 0.2653
	DaDeblur [12]	29.54 / 0.8772	26.8422 / 0.2025	32.3833 / 0.2322	25.3244 / 0.2684
	SelfHVD _{IFIRNN}	34.32 / 0.9302	27.6922 / 0.2137	32.7765 / 0.2530	25.7711 / 0.2756
	SelfHVD _{ESTRNN}	33.60 / 0.9216	27.6873 / 0.2126	32.4002 / 0.2574	25.7556 / 0.2842
	SelfHVD _{RVRT}	36.31 / 0.9300	27.8345 / 0.2088	32.4416 / 0.2303	25.8437 / 0.2711
	SelfHVD _{BasicVSR++}	37.44 / 0.9359	28.0040 / 0.2175	32.8564 / 0.2236	25.7022 / 0.2686

Table 2. Quantitative results on BSD [64], RealBlur [40] and RBVD [4] via test-time fine-tuning. Each group reports PSNR and SSIM. All methods use ESTRNN [64] as the deblurring network, with its GoPro-trained version serving as the baseline.

Methods	BSD-1ms8ms [64]	BSD-2ms16ms [64]	BSD-3ms24ms [64]	RealBlur [40]	RBVD [4]
	PSNR↑ / SSIM↑	PSNR↑ / SSIM↑	PSNR↑ / SSIM↑	PSNR↑ / SSIM↑	PSNR↑ / SSIM↑
Baseline	25.57 / 0.747	24.64 / 0.726	26.01 / 0.748	25.87 / 0.773	24.47 / 0.725
+Blur2Blur [39]	25.64 / 0.750	24.72 / 0.728	26.01 / 0.749	26.16 / 0.800	24.45 / 0.725
+DaDeblur [12]	29.44 / 0.843	28.36 / 0.820	28.23 / 0.808	27.41 / 0.819	27.02 / 0.771
+Ours	31.01 / 0.873	29.00 / 0.832	29.31 / 0.825	28.76 / 0.848	27.69 / 0.785

■ 手机拍摄视频自监督去模糊方法SelfHVD

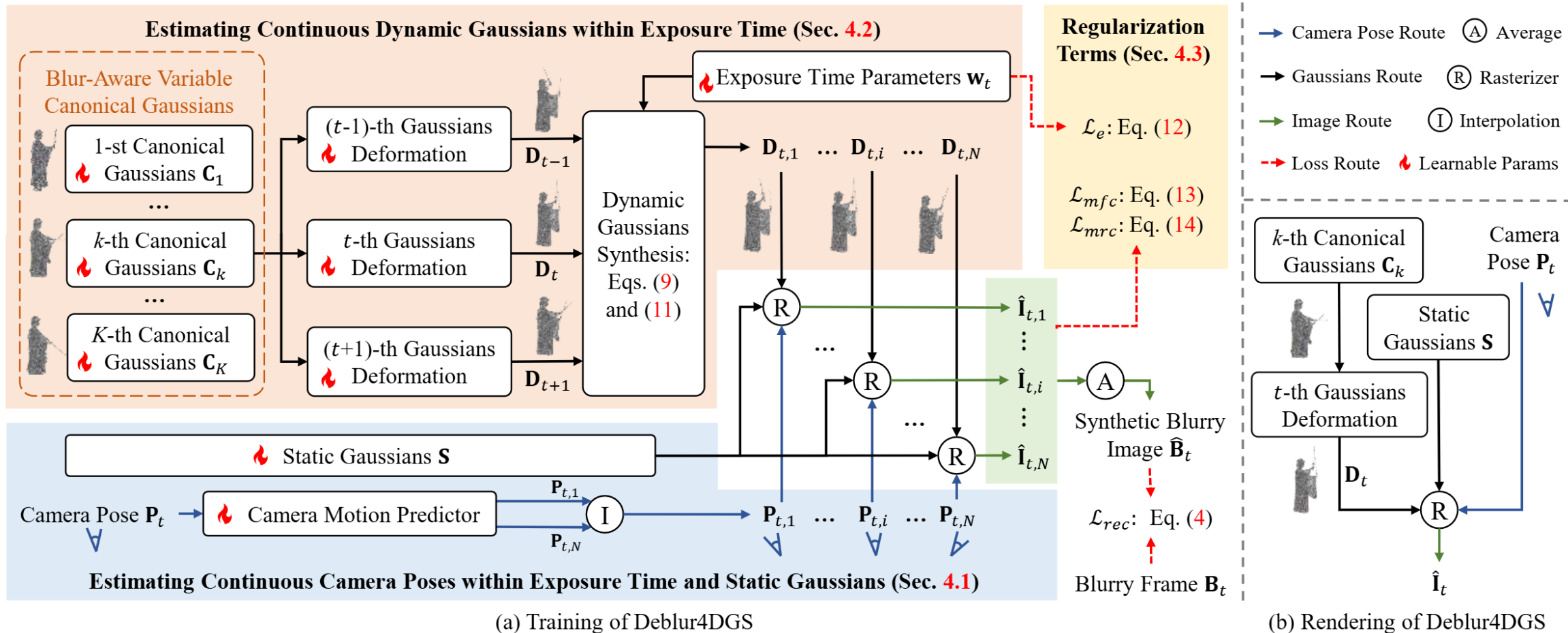


输入视频



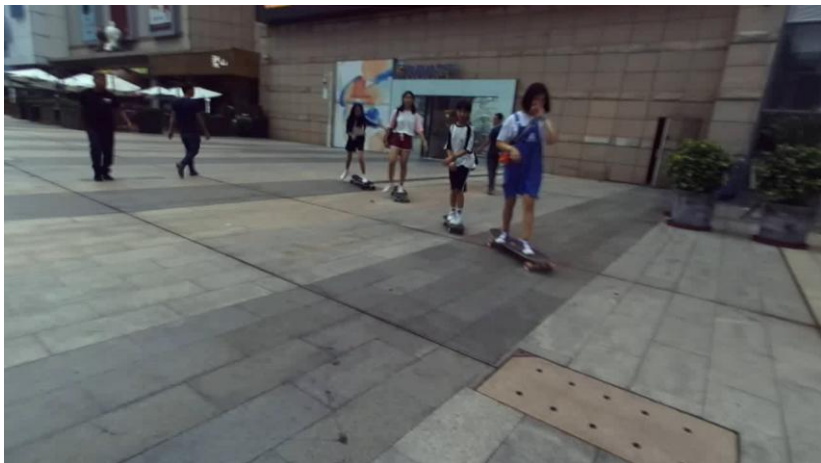
输出视频

■ 单目模糊视频4D重建方法Deblur4DGS



■ 单目模糊视频4D重建方法Deblur4DGS

原模糊
视频



插帧



新视角
生成



新视角
生成



■ 单目模糊视频4D重建方法Deblur4DGS

原模糊
视频



视频增稳
结果





三、多曝光图像复原

■ 多曝光图像

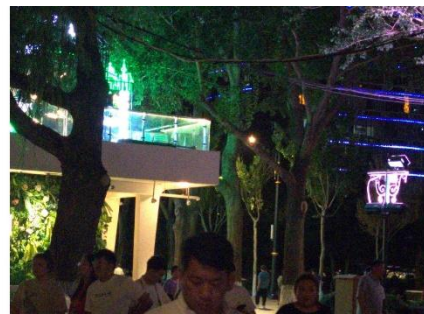
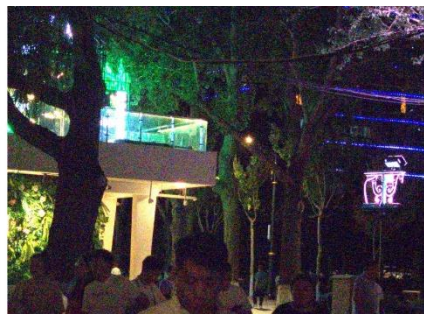
短曝光



低信噪比

锐利

欠曝



长曝光



高信噪比

模糊

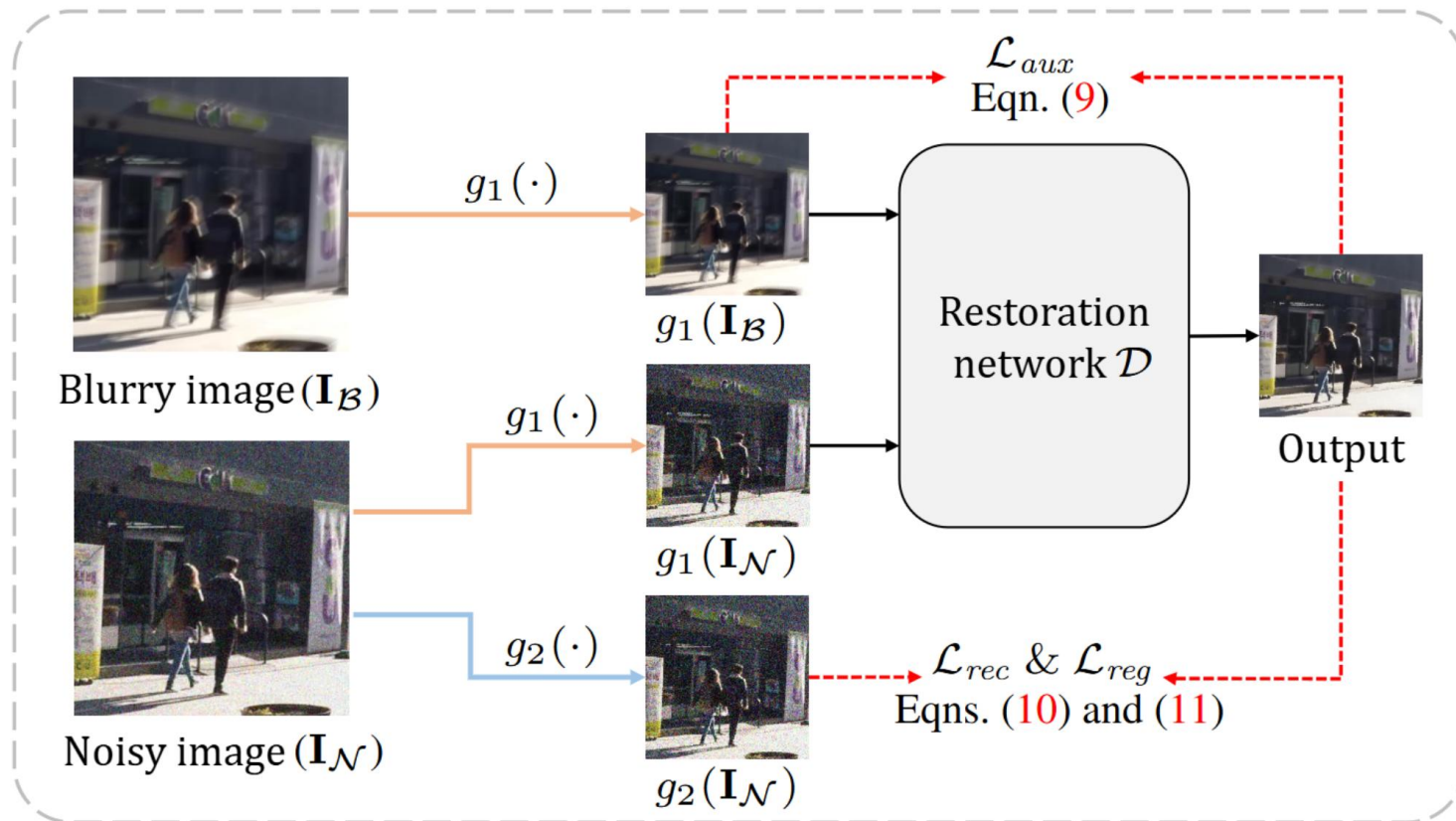
过曝

帧间的空间位置不对齐问题

■ SelfIR: 考虑双曝图像中的模糊和噪声

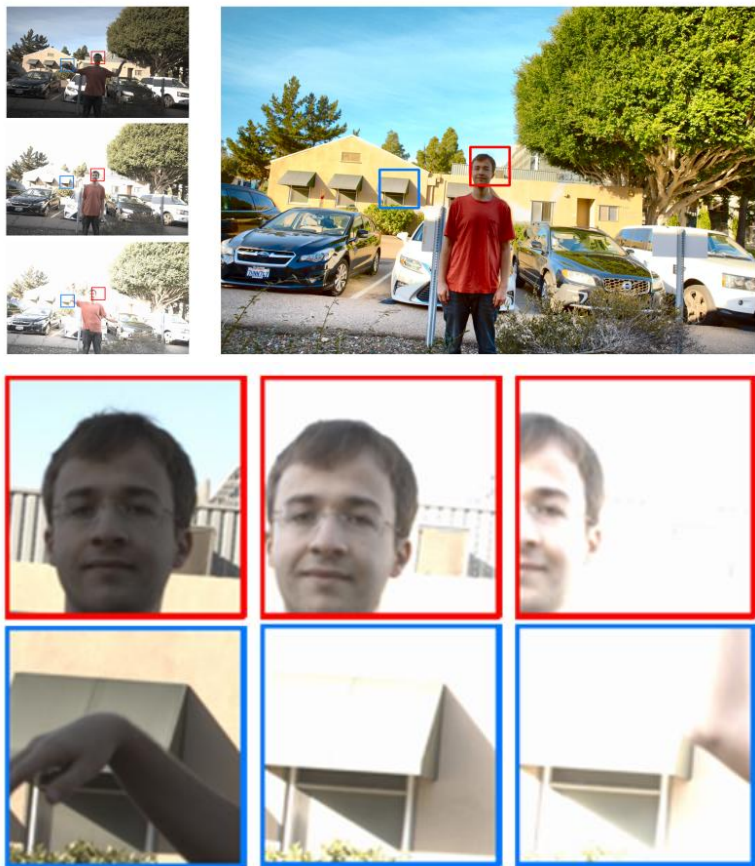
□ 噪声图像可为去模糊提供监督

□ 长曝图像的清晰区域可为图像去噪提供监督

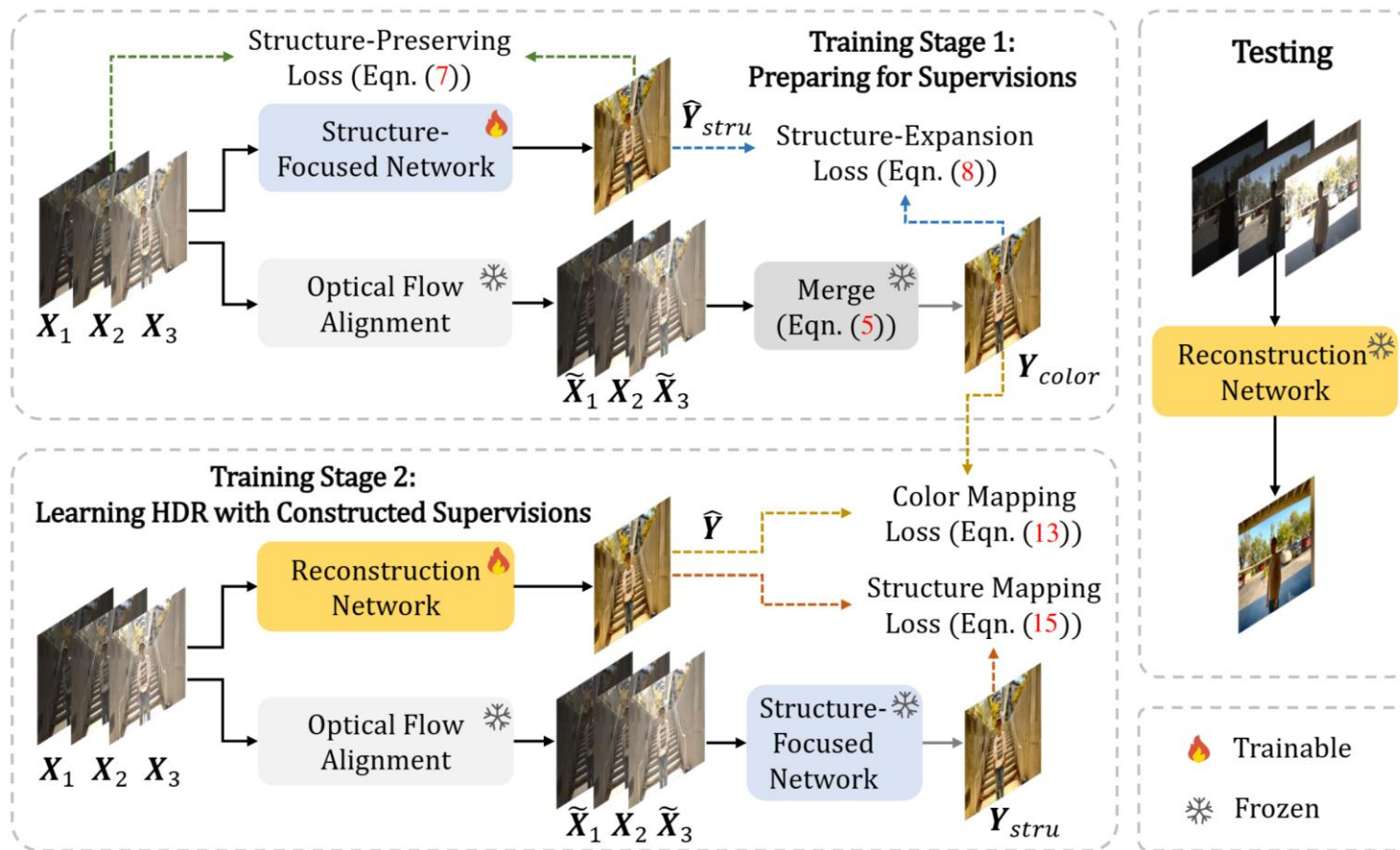


$$\Theta_D^* = \arg \min_{\Theta_D} \mathbb{E}_{\mathbf{I}_B, \mathbf{I}_N} (\mathcal{L}_{rec} + \lambda_{reg} \mathcal{L}_{reg} + \lambda_{aux} \mathcal{L}_{aux})$$

SelfHDR: 考虑白天场景的HDR



挑战



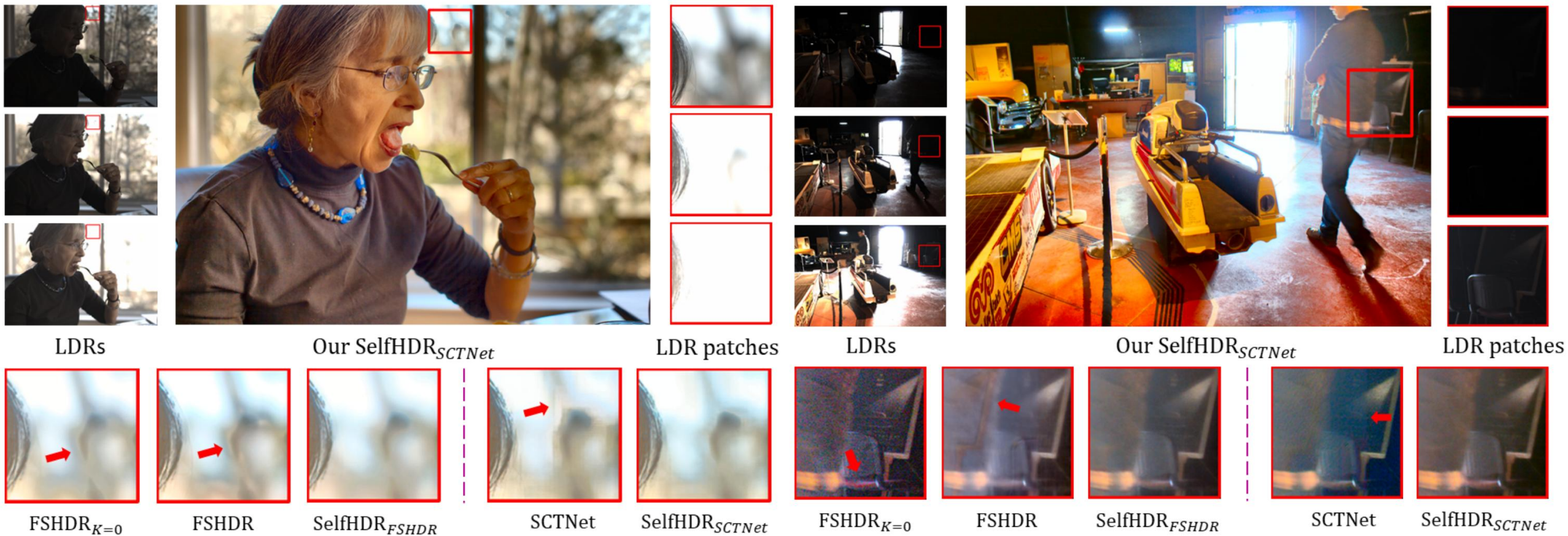
两阶段学习方法

SelfHDR: 考虑白天场景的HDR

	Method	PSNR- u / SSIM- u	PSNR- l / SSIM- l	HDR-VDP-2
Fully-Supervised	AHDRNet (CVPR 2019)	43.63 / 0.9900	41.14 / 0.9702	64.61
	FSHDR (CVPR 2021)	43.03 / 0.9902	42.27 / 0.9889	64.79
	HDR-Transformer (ECCV 2022)	44.21 / 0.9918	42.17 / 0.9889	64.63
	SCTNet (ICCV 2023)	44.48 / 0.9916	42.00 / 0.9897	64.47
Few (K)-Shot	FSHDR $_{K=5}$ (CVPR 2021)	43.02 / 0.9874	41.98 / 0.9885	64.54
	FSHDR $_{K=1}$ (CVPR 2021)	42.52 / 0.9846	41.92 / 0.9887	64.41
Self-Supervised	FSHDR $_{K=0}$ (CVPR 2021)	42.17 / 0.9828	41.47 / 0.9884	64.21
	Nazarczuk <i>et al.</i> (ArXiv 2022)	42.15 / -	40.54 / -	63.99
	Our SelfHDR $_{AHDRNet}$	43.68 / 0.9901	41.09 / 0.9873	64.57
	Our SelfHDR $_{FSHDR}$	43.80 / 0.9902	41.72 / 0.9880	64.57
	Our SelfHDR $_{HDR-Transformer}$	43.94 / 0.9907	41.79 / 0.9883	64.98
	Our SelfHDR $_{SCTNet}$	43.95 / 0.9907	41.77 / 0.9889	64.77

自监督HDR方法可以取得与全监督相当或更好的性能

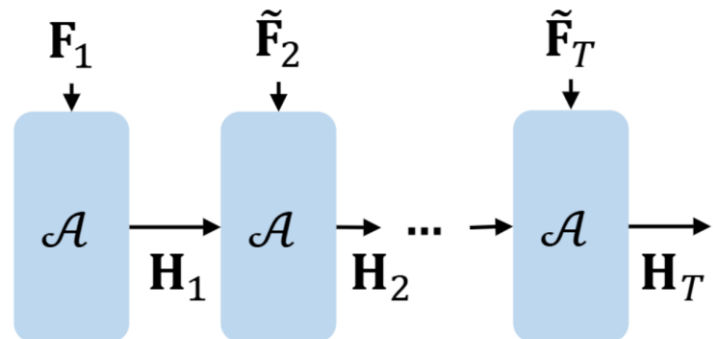
SelfHDR: 考虑白天场景的HDR



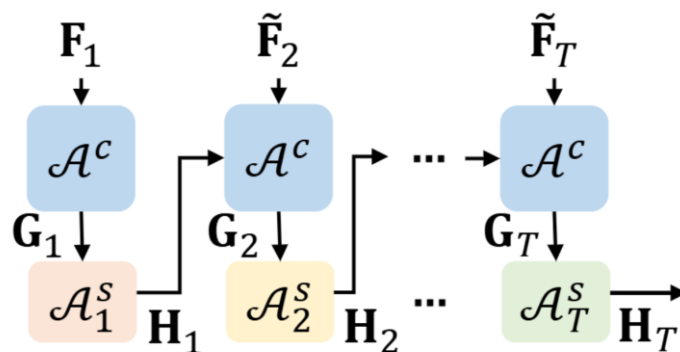
自监督HDR方法可以取得与全监督相当或更好的性能

- BracketIRE: 将常见图像复原和增强任务（去噪、去模糊、超分、HDR）纳入到一个统一框架

网络

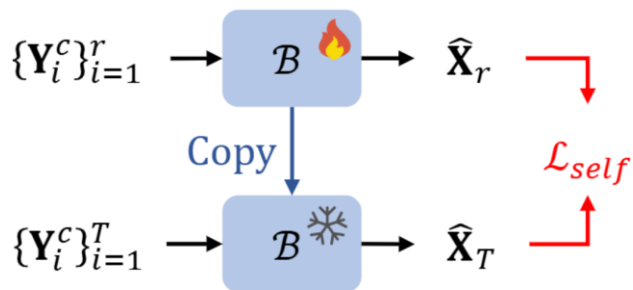


(a) Baseline Recurrent Network

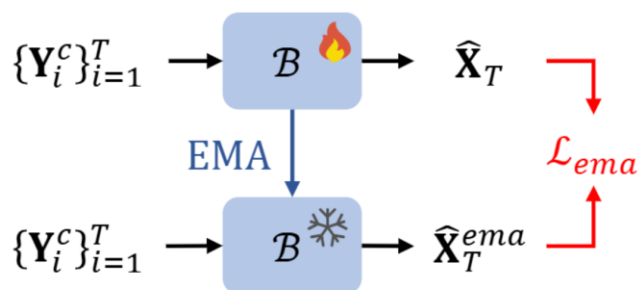


(b) Our TMRNet

自监督
自适应
微调



(a) Temporally Self-Supervised Loss



(b) EMA Regularization Loss

$$\mathcal{L}_{self} = \|\mathcal{T}(\hat{\mathbf{X}}_r) - \mathcal{T}(sg(\hat{\mathbf{X}}_T))\|_1$$

$$\mathcal{L}_{neg} = -\|\mathcal{T}(\hat{\mathbf{X}}_T) - \mathcal{T}(sg(\hat{\mathbf{X}}_{T-1}))\|_1$$

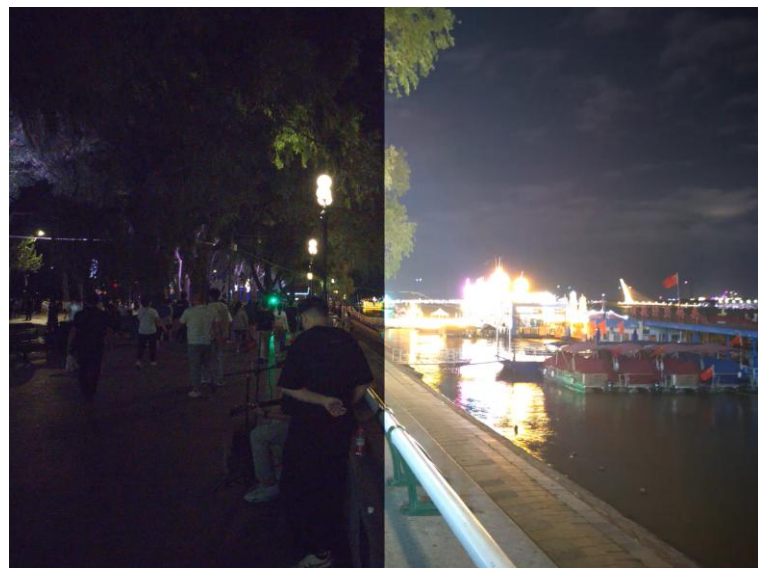
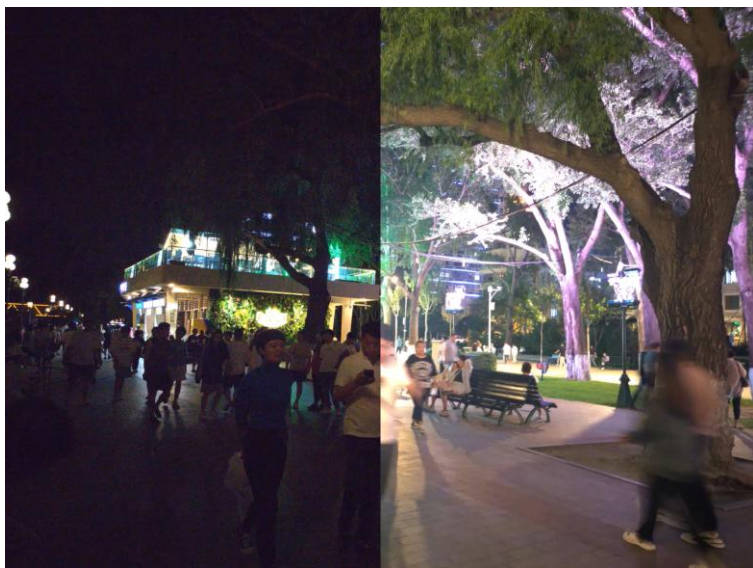
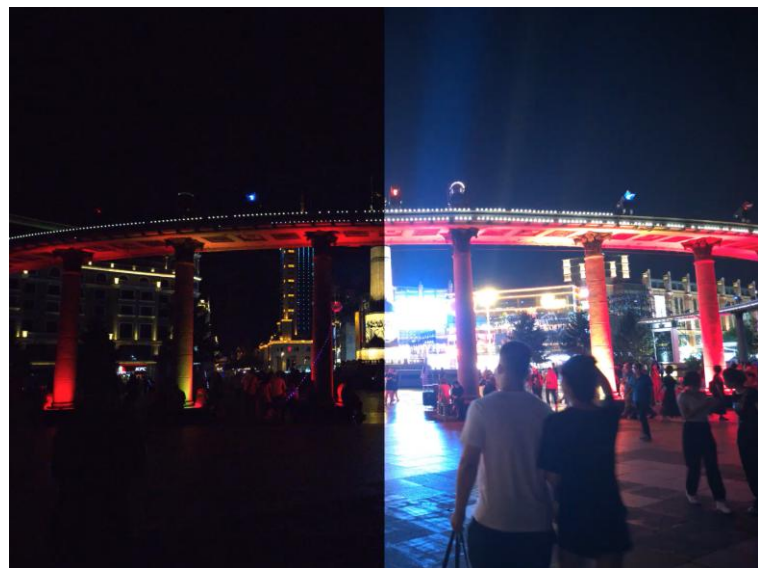
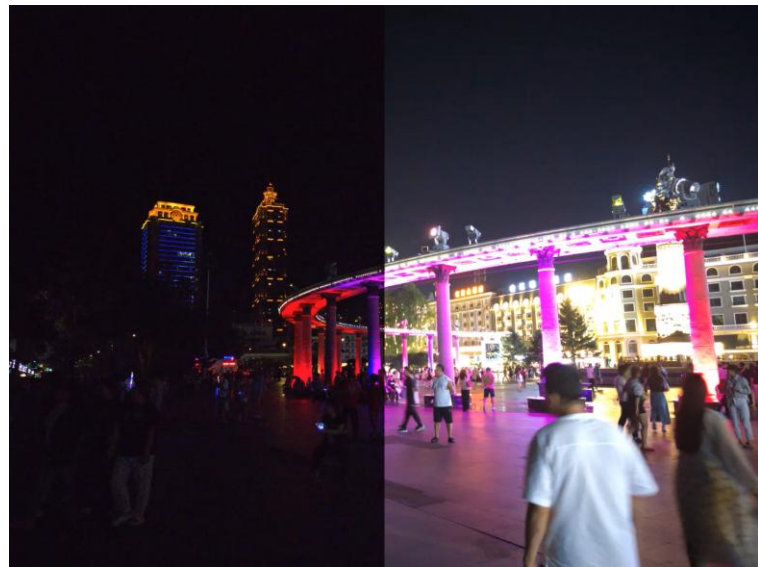
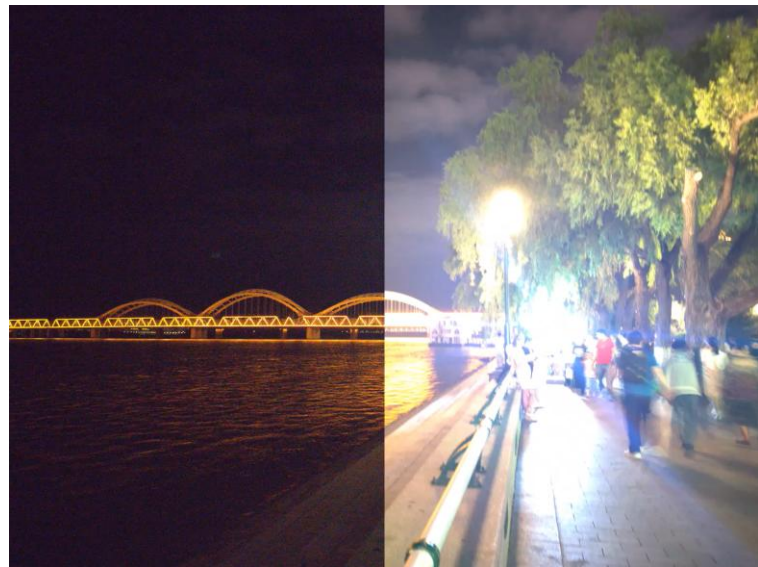
$$\mathcal{L}_{ema} = \|\mathcal{T}(\hat{\mathbf{X}}_T) - \mathcal{T}(sg(\hat{\mathbf{X}}_T^{ema}))\|_1$$

■ BracketIRE: 将常见图像复原和增强任务（去噪、去模糊、超分、HDR）
纳入到一个统一框架

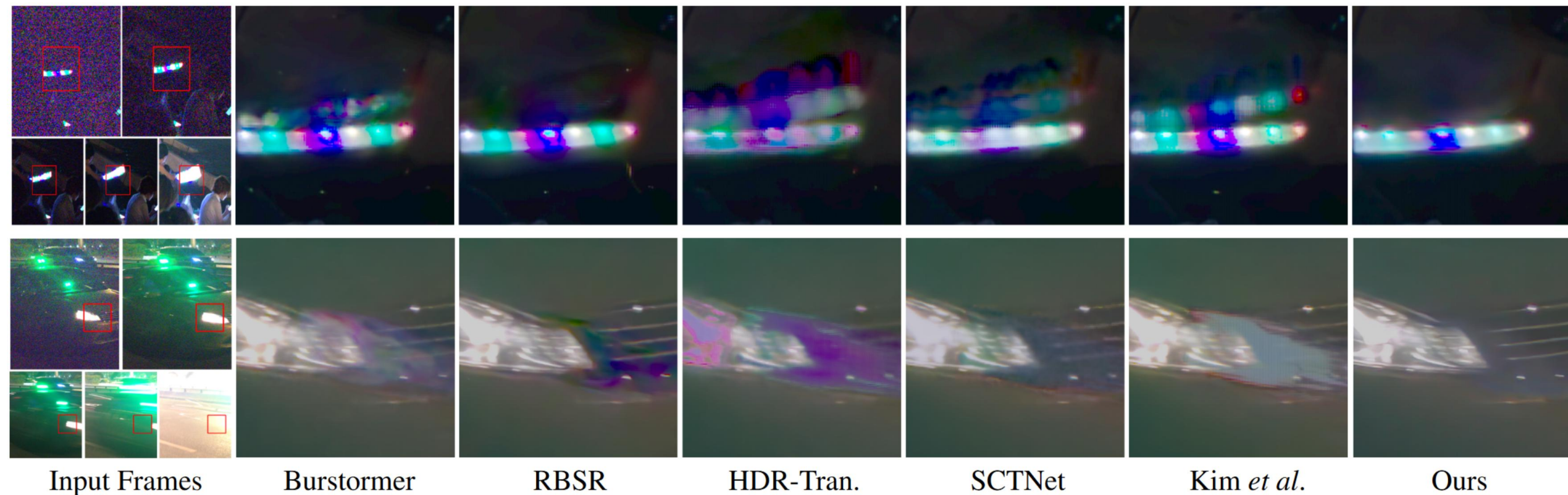
	Method	BracketIRE		BracketIRE+	
		Synthetic	Real-World	Synthetic	Real-World
		PSNR↑/SSIM↑/LPIPS↓	CLIPQA↑/MANIQA↑	PSNR↑/SSIM↑/LPIPS↓	CLIPQA↑/MANIQA↑
Burst Processing Networks	DBSR	35.13/0.9092/0.188	0.1359/0.1653	29.79/0.8546/0.335	0.3340/0.2911
	MFIR	35.64/0.9161/0.177	0.2192/0.2310	30.06/0.8591/0.319	0.3402/0.2908
	BIPNet	36.92/0.9331/0.148	0.2234/0.2348	30.02/0.8582/0.324	<u>0.3577</u> /0.2979
	Burstormer	37.06/0.9344/0.151	0.2399/ <u>0.2390</u>	29.99/0.8617/0.300	<u>0.3549</u> / <u>0.3060</u>
	RBSR	<u>39.10</u> / <u>0.9498</u> /0.117	0.2074/0.2341	<u>30.49</u> / <u>0.8713</u> /0.275	0.3425/0.2895
HDR Reconstruction Networks	AHDRNet	36.68/0.9279/0.158	0.2010/0.2259	29.86/0.8589/0.308	0.3382/0.2909
	HDRGAN	35.94/0.9177/0.181	0.1995/0.2178	30.00/0.8590/0.337	0.3555/ 0.3109
	HDR-Tran.	37.62/0.9356/0.129	0.2043/0.2142	30.18/0.8662/0.279	0.3245/0.2933
	SCTNet	37.47/0.9443/0.122	0.2348/0.2260	30.13/0.8644/0.281	0.3415/0.2936
	Kim <i>et al.</i>	39.09/0.9494/ <u>0.115</u>	<u>0.2467</u> /0.2388	30.28/0.8658/ 0.268	0.3302/0.2954
Our TMRNet	w/o Ada.	39.35/0.9516/0.112	0.2003/0.2181	30.65/0.8725 / <u>0.270</u>	0.3422/0.2898
	w/ Ada.	-	0.2537/0.2422	-	0.3676 /0.3020

多曝图像融合复原

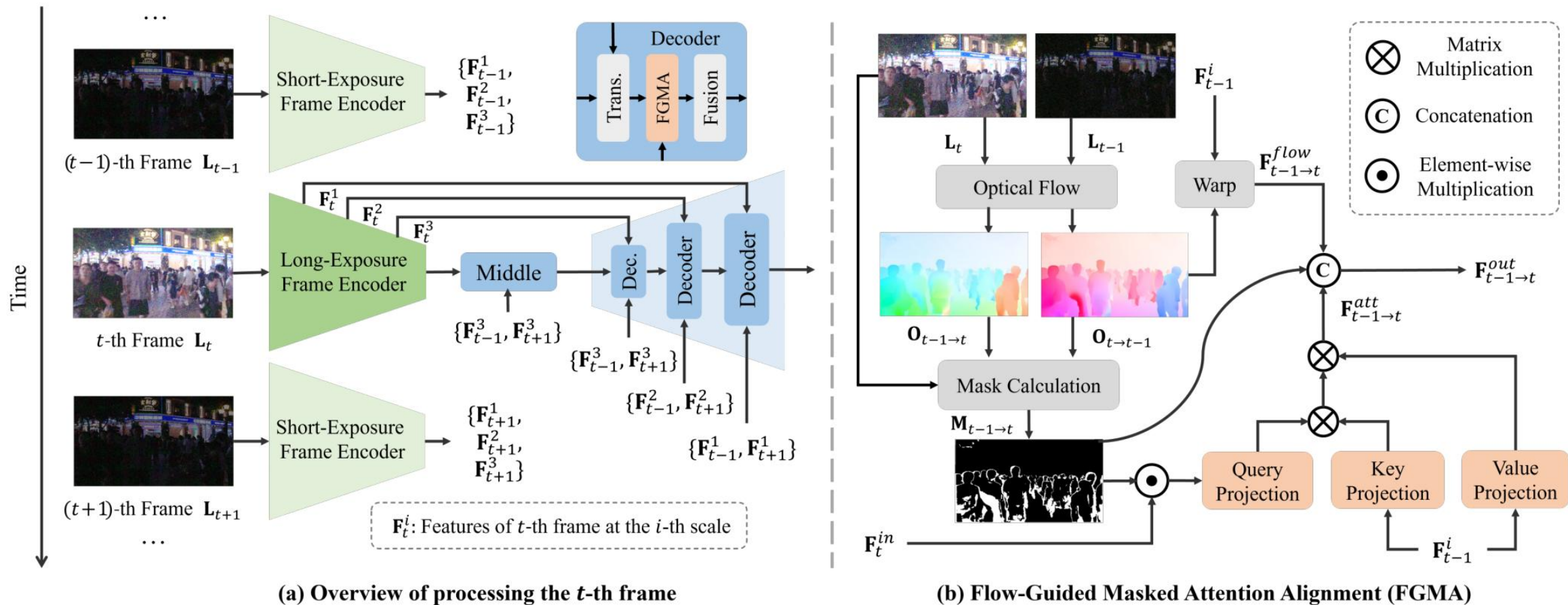
去噪&去模糊&HDR&超分



- BracketIRE: 将常见图像复原和增强任务（去噪、去模糊、超分、HDR）纳入到一个统一框架



多曝视频复原方法：DeAltHDR



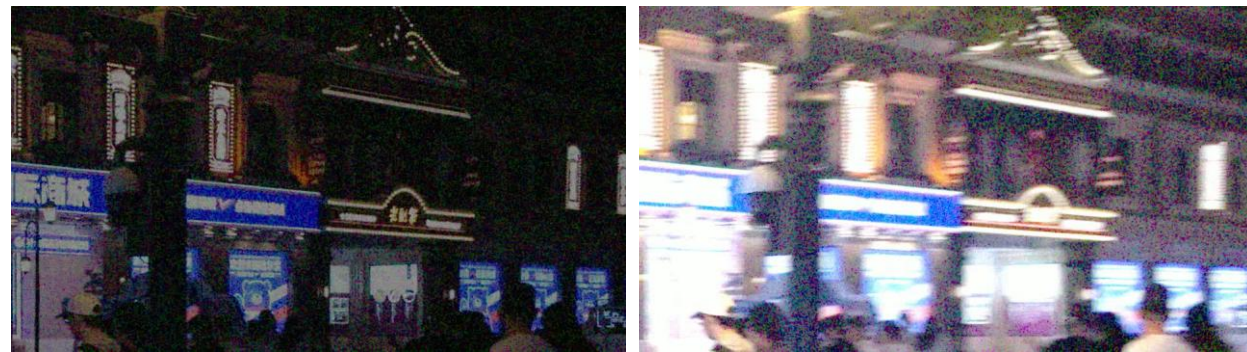
多曝图像融合复原

多曝视频复原：去噪+去模糊+HDR

■ 多曝视频复原方法：DeAltHDR



双曝光输入



双曝光输入



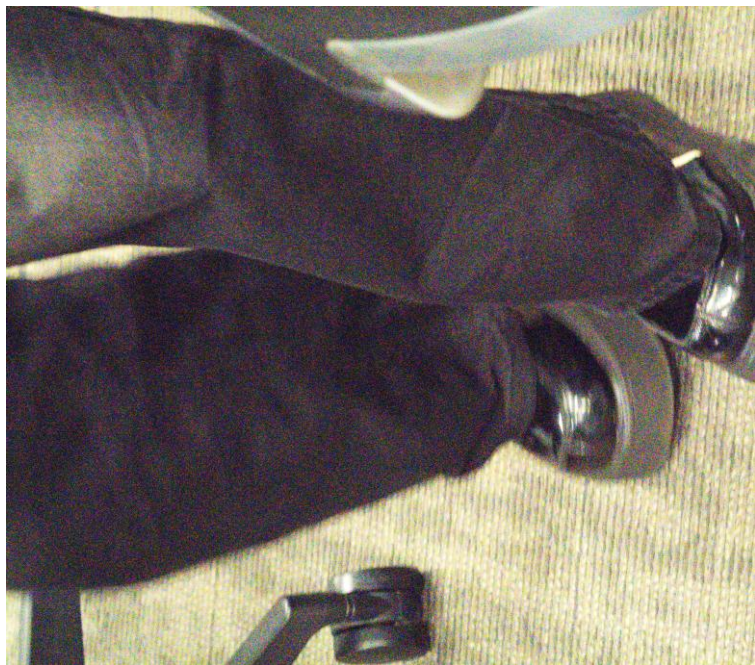
复原
结果





四、端側落地情況

■ 多帧去噪：抑制低频彩噪



噪声图



手机商用模型结果



自监督微调模型结果

■ 多帧去噪：抑制低频彩噪



噪声图



手机商用模型结果



自监督微调模型结果

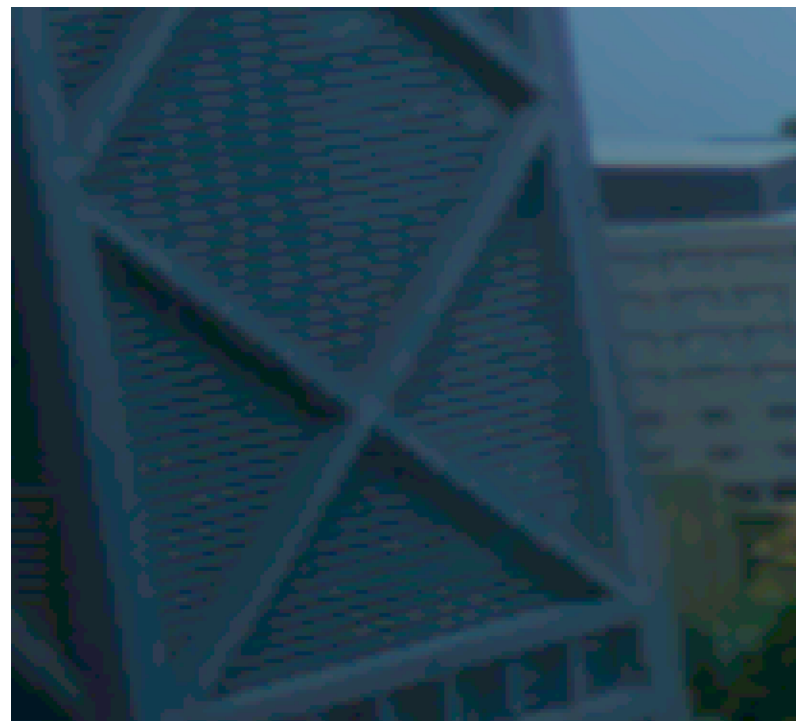
■ 视频去噪：增强细节



噪声图



手机商用模型结果



自监督微调模型结果

■ 视频去噪：增强细节



噪声图



手机商用模型结果



自监督微调模型结果

■ 视频去模糊：增强细节



模糊图



手机商用模型结果



自监督微调模型结果

■ 视频去模糊：增强细节



模糊图



手机商用模型结果



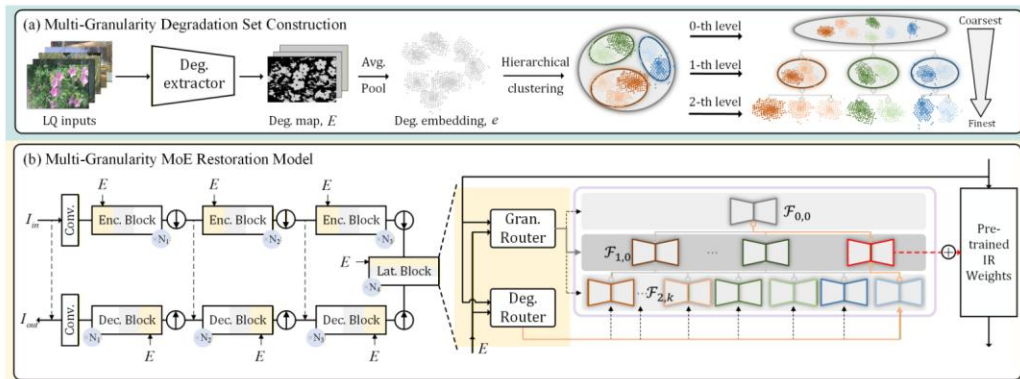
自监督微调模型结果



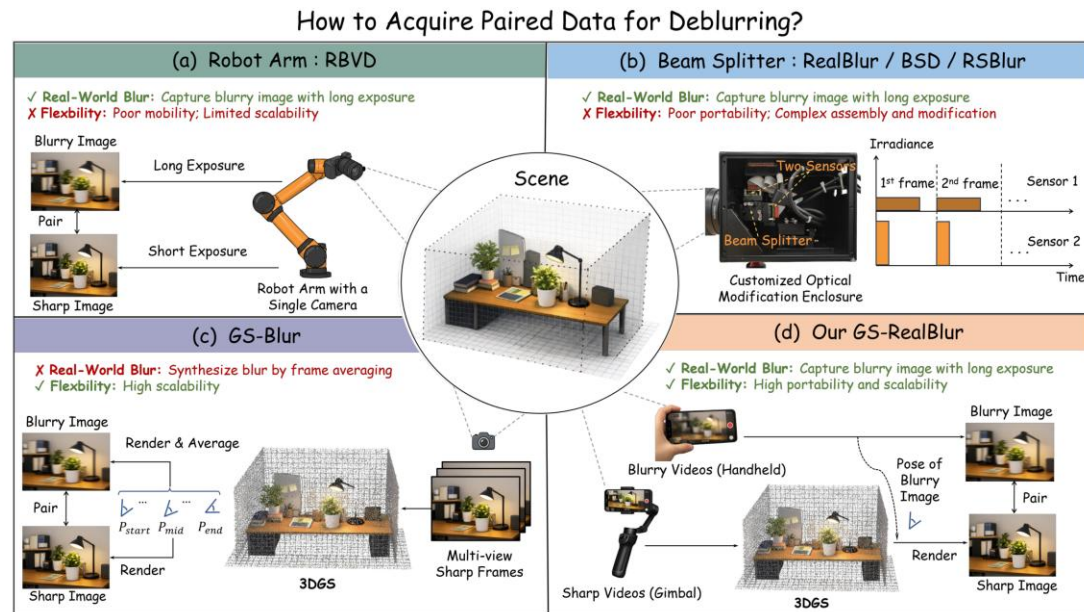
总结与展望

总结与展望

- 自监督是数据之殇下的无奈or必须?
- 高质量训练数据是关键——宁缺毋滥
 - 数量、场景多样性
 - 仿真数据：退化的真实性和分布
 - 真实数据：配对数据的严格对齐
- 更加高效、灵活地收集、构造、扩展数据
- 数据完备时的展望：有效实现复原模型的Scaling Up



总结与展望



Method	SIDD [79]	RealBlur-J [80]	DPDD [81]	RealRain [82]	RealSnow [32]	FoundIR [83]
NAFNet [11]	22.82	25.33	20.76	29.76	27.51	21.11
Restormer [10]	23.06	26.35	22.76	29.90	26.73	21.45
PromptIR [17]	24.32	25.47	21.76	24.12	24.94	21.54
MiOIR [20]	22.87	24.51	20.41	26.68	26.17	23.27
OneRestore [19]	11.46	18.79	20.96	22.73	20.72	22.94
MoCEIR [20]	15.96	14.29	15.61	28.10	24.63	22.63
UniRestorer [36]	24.70	25.55	23.60	29.85	27.73	23.80
UniRestorer++	24.85	26.25	23.90	29.95	27.80	24.05
UniRestorer++ (3B)	38.72	30.81	24.91	31.37	28.88	28.17

相关论文

相关论文

1. **Zhilu Zhang**, RongJian Xu, Ming Liu, Zifei Yan, and Wangmeng Zuo; “Self-Supervised Image Restoration with Blurry and Noisy Pairs”, **NeurIPS 2022**.
2. **Zhilu Zhang**, Haoyu Wang, Shuai Liu, Xiaotao Wang, Lei Lei, and Wangmeng Zuo; “Self-Supervised High Dynamic Range Imaging with Multi-Exposure Images in Dynamic Scenes”, **ICLR 2024**.
3. **Zhilu Zhang**, Shuohao Zhang, Renlong Wu, Zifei Yan, Wangmeng Zuo; “Exposure Bracketing Is All You Need for a High-quality Image”, **ICLR 2025**.
4. Renlong Wu, **Zhilu Zhang***, Shuohao Zhang, Longfei Gou, Haobin Chen, Lei Zhang, Hao Chen*, Wangmeng Zuo; “Self-Supervised Video Desmoking for Laparoscopic Surgery”, **ECCV 2024 (Oral)**.
5. Renlong Wu, **Zhilu Zhang***, Mingyang Chen, Zifei Yan, Wangmeng Zuo; “Deblur4DGS: 4D Gaussian Splatting from Blurry Monocular Videos”, **AAAI 2026**.
6. Shuohao Zhang, **Zhilu Zhang***, RongJian Xu, Xiaohe Wu, Wangmeng Zuo; “DeAltHDR: Learning HDR Video Reconstruction from Degraded Alternating Exposure Sequences”, **ICLR 2026**.
7. Honglei Xu, **Zhilu Zhang***, Junjie Fan, Xiaohe Wu, Wangmeng Zuo; “SelfHVD: Self-Supervised Handheld Video Deblurring”, **CVPR 2026**.
8. Junyi Li, **Zhilu Zhang**, Xiaoyu Liu, Chaoyu Feng, Xiaotao Wang, Lei Lei, Wangmeng Zuo*; “Spatially Adaptive Self-Supervised Learning for Real-World Image Denoising”, **CVPR 2023**.
9. Junyi Li, **Zhilu Zhang**, Wangmeng Zuo*; “Rethinking Transformer-Based Blind-Spot Network for Self-Supervised Image Denoising”, **AAAI 2025**.

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致谢



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致谢



李俊义



吴仁龙



徐洪磊



张朔豪



**谢谢！
敬请指导交流！**

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